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Review

Advancements in UAV-Enabled Intelligent Transportation Systems: A Three-Layered Framework and Future Directions

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Featured Application: This paper addresses the practical application of unmanned aerial vehicles (UAVs) in modern transportation systems, mainly focusing on how UAVs can be effectively integrated into intelligent transportation systems (ITSs) to improve efficiency, safety, and decision-making in smart cities. It offers a framework for understanding current advancements and guides future research and development in this evolving field.

Abstract: Integrating unmanned aerial vehicles (UAVs) into intelligent transportation systems (ITSs) will be pivotal in shaping next-generation smart cities. This paper proposes a novel three-layered framework for integrating UAVs into intelligent transportation systems (ITSs) and reviews the current developments, challenges, and future directions in this emerging field. This framework provides a comprehensive overview of the key components of UAV-integrated ITSs, encompassing UAV specifications and deployment strategies, communication networks, and data utilization for traffic management. The first layer explores UAVs' technical specifications, deployment strategies, and trajectory optimization, essential for maximizing UAV performance in transportation contexts. The second layer addresses the communication networks between UAVs and vehicles, along with the use of UAVs for responsive traffic monitoring. This includes the development of robust communication protocols and real-time traffic analysis to enhance system efficiency. The third layer focuses on advanced data collection processing techniques and complexities, reviewing the methods for analyzing the traffic data collected by UAVs for decision-making in transportation management. Moreover, the paper presents the current UAV-enabled ITS implementation, highlighting key challenges and future research directions. By providing a comprehensive overview of UAV-enabled ITSs, this study presents a significant portrayal of the current landscape of UAV integration in ITSs and serves as a foundation for future advancements in smart city infrastructure.

Keywords: unmanned aerial vehicles (UAV); intelligent transportation systems (ITS); communication; networks; data processing

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1. Introduction

The term "smart cities" is an emerging concept of modern civilization that uses technology to improve transportation, accessibility, and social services and to promote sustainability to provide a more manageable life for citizens [1]. One of the fundamental elements of a smart city is the intelligent transportation system (ITS), which uses a collection of connected technologies to run an automated traffic management system [2]. An ITS collects and analyzes data to make predictive and adaptive decisions to enhance functional efficiency. The continuously evolving ITS technologies have benefited traffic management systems [3]. However, the deployment of ITS technologies has not yet been widely implemented in all possible applications. Upcoming technologies, such as connected autonomous vehicles and unmanned aerial vehicles (UAVs), are crucial components of the large-scale, global deployment of ITSs [4,5]. Their integration can significantly improve traffic management by alleviating congestion and enhancing traffic flow [6].

Traffic congestion increases yearly with the increasing number of cars and insufficient road capacity, which has social and economic impacts [7]. In 2019, American drivers lost an average of 99 h annually due to traffic congestion, costing approximately USD 1377 per person [7]. By 2020, the average time lost was reduced to 26 h, primarily due to the COVID-19 pandemic, which translated into a savings of USD 51 billion or about USD 980 per driver [8]. This suggests that alleviating traffic congestion will reduce time lost and associated costs, enhancing social and economic conditions. However, more recent data from 2022 indicate that congestion has begun to return to pre-pandemic levels, with drivers experiencing an average of 51 h of delay, costing around USD 1300 per driver [9]. Therefore, the primary goal of ITSs is to ensure efficient traffic flow, improve transportation safety, and enhance productivity through the implementation of advanced information technologies [10].

To achieve this goal, UAVs can be integrated into the transportation system, including tracking and monitoring ground vehicles, supporting logistics by replacing conventional methods, and providing real-time emergency response with autonomous UAVs. Many logistics companies have recently adopted UAV-based transportation to bypass traffic congestion and ensure faster deliveries. The global market for UAV-assisted logistics and transportation is projected to grow significantly from USD 5.3 billion in 2019 to USD 11 billion by 2026 [11]. Additionally, UAVs can be utilized in other transportation-related areas, including traffic monitoring, urban planning, and disaster response. As the number of automated vehicles increases, UAVs will play a crucial role in advancing computerized transportation systems [4]. Ongoing developments in this field are opening up new research opportunities and applications, with many researchers and transportation professionals actively working towards establishing UAV-enabled ITSs [5,12].

The benefits of UAV-enabled ITSs have been mainly described in Ref. [13]. The technical contribution of UAV integration into ITSs offers several advantages. UAVs equipped with cameras and sensors can monitor traffic from an aerial perspective, complementing ground-based sensors and cameras. They provide real-time traffic updates, detect congestion, and monitor accidents more efficiently in areas that are otherwise hard to cover using traditional ITS infrastructure like road cameras or sensors. UAVs can quickly reach incident sites, providing high-resolution imagery and live video feeds to ITS control centers. This supports quicker detection of and response to road accidents, natural disasters, or infrastructure damage. UAVs collect data from large geographical areas, feeding them into ITS databases for advanced analytics. These data include traffic flow, weather conditions, and infrastructure health. UAVs also enhance communication by acting as aerial relays in areas where ground-based communication networks are sparse or temporarily disabled. UAVs are used for environmental monitoring tasks like air quality checks and noise level assessments, integrating these data into ITSs to maintain sustainable transportation networks.

However, despite these advantages, significant limitations must be addressed before large-scale implementation occurs. To provide a holistic view of the complexities and opportunities in developing a UAV-based transportation system, a three-layered framework is proposed to analyze the interdependencies among the layers, along with the cybersecurity measures and expertise required to build a smart transportation system. Key issues include UAV security and privacy concerns, challenges with automated UAV navigation, and the limited battery life of current UAVs [14,15]. To successfully integrate UAVs into future transportation systems, it is crucial to have a comprehensive understanding of their advancements, challenges, and applications in this sector. Although UAVs have been explored in various domains, their full potential in transportation remains underexplored. Existing research often focuses on specific aspects rather than providing a holistic overview. Therefore, this paper comprehensively reviews recent developments in UAVs within transportation systems using a structured three-layered framework. The framework addresses critical aspects of UAV integration into ITSs, including (1) UAV technical specifications, (2) communication networks for responsive traffic monitoring, and (3) advancements in data collection and processing. This framework bridges the gap between theoretical advancements and practical applications by demonstrating how the integration of UAV technologies can enhance transportation systems. Moreover, it identifies future research and development opportunities in this field.

Although the authors propose a three-layered framework, it ideally serves as a structured overview of previous and current fragmented efforts organized within each subtopic or layer toward a successful UAV-enabled ITS application. Organizing the content in this manner provides a comprehensive perspective on the advancements in UAV-enabled ITSs, guiding future research directions while acknowledging the historical context that has shaped the field. The objectives of this review paper are as follows: (1) to present a detailed summary of UAV-enabled ITSs within the three-layered framework, (2) to analyze the current state of each aspect of the framework, and (3) to identify future research directions for developing robust UAV-enabled ITSs. The rest of the paper is organized as follows. Section 2 proposes examining UAV-enabled ITSs under a three-layered framework. The results and analysis of the current fragmented efforts related to the proposed framework are discussed separately for each layer in Sections 2.1–2.3. Additionally, the discussion is separated into two parts, with Section 3 exploring the current research areas and application of UAVs in transportation and Section 4 examining the challenges, opportunities, and future research directions in the realm of UAV-enabled ITSs. Finally, in Section 5, an overview of the paper is presented.

2. UAV-Enabled ITS Framework

Future UAV-enabled ITSs could have various critical components significant for their successful implementation. To properly understand the step-by-step implementation of UAV-enabled ITSs, crucial components will be discussed using the three-layered framework shown in Figure 1.

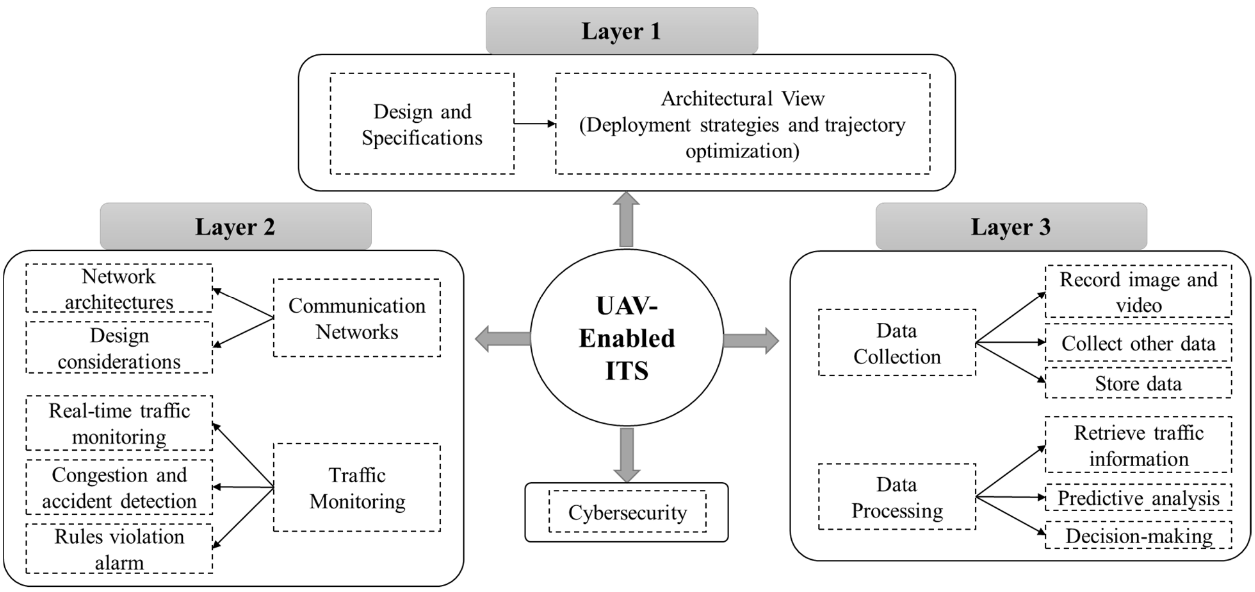

Figure 1. UAV-enabled intelligent transportation system (ITS).

The first layer of the UAV-enabled ITS focuses on UAVs' technical and operational properties. It further involves the following three key factors: (1) defining UAV design and specifications to meet specific requirements, (2) optimizing UAV deployment to ensure maximum coverage and efficiency, and (3) analyzing UAV trajectory regulations for effective traffic monitoring.

The second layer encompasses two main components. The first component is the communication networks essential to the ITS infrastructure, including vehicle-to-vehicle (V2V), vehicle-to-UAV (V2U), UAV-to-UAV (U2U), and UAV-to-vehicle (U2V) communications. These networks should also be connected to the Internet and cloud services for efficient data management and regulation. The second component is the responsive traffic monitoring system, which continuously monitors traffic flow and detects anomalies such as accidents, roadwork, adverse weather conditions, and rule violations [16]. The system is designed to respond promptly to these events to maintain optimal traffic conditions.

The third layer of the framework focuses on data collection and processing. In addition to visual traffic data, other data types, such as infrared, light detection and ranging (LiDAR), radar, acoustic, and environmental data, can be collected using sensors and then transmitted to storage devices or uploaded to cloud servers. These data are processed to extract valuable traffic information that can provide comprehensive situational traffic awareness. The processed information can be used for further predictive analysis and decision-making to improve overall traffic flow.

Additionally, cybersecurity must be integrated at every layer in this three-layer UAV-enabled ITS framework to ensure overall system integrity and safety. At Layer 1, technical specifications should include robust security protocols to protect the UAV's hardware and software from vulnerabilities. Layer 2, which encompasses the communication network, requires secure channels to safeguard data transmission between UAVs and ground control stations against interception or attacks. Finally, Layer 3 focuses on data collection and processing, where implementing stringent cybersecurity measures is essential to protect sensitive information and ensure the accuracy and reliability of data analysis. This comprehensive approach to cybersecurity helps create a resilient UAV-enabled ITS framework.

Introducing the three-layered framework for UAV-enabled ITSs significantly advances existing research. This innovative approach provides a structured approach to the UAV-enabled ITS implementation process and offers holistic solutions that enhance scalability, efficiency, and real-time performance in ways that previous fragmented efforts have often struggled to achieve. Regarding scalability, the first layer discusses the UAV specification and requirements, facilitating a scalable model that can adapt to various urban environments and traffic scenarios. The second layer emphasizes efficient communication, directly addressing the limitations of conventional systems that frequently encounter delays and bandwidth constraints. Additionally, existing research often overlooks the complexities of real-time data analysis, leading to delays in decision-making. Therefore, the third layer of the framework should prioritize real-time processing capabilities, ensuring that traffic data can be swiftly analyzed and utilized for immediate decision-making. The three-layered framework serves as a structural means to cover various significant elements of the ITS.

2.1. First Layer: Technical Specifications

The first layer of the UAV-enabled ITS framework focuses on the technical specifications of UAVs. It includes defining design parameters, optimizing deployment for maximum coverage, and regulating flight trajectories for effective traffic monitoring. These elements are essential for ensuring UAVs meet the operational needs of ITSs.

2.1.1. Design and Specifications

For the successful integration of UAVs into ITSs, the design and specifications of UAVs should adhere to the requirements. UAVs are categorized by their lift mechanism into three main types, fixed-wing, rotary-wing, and hybrid [14,17], as depicted in Figure 2. Fixed-wing UAVs typically use propellers to provide horizontal thrust, making them more energy-efficient and able to achieve longer flight times than multi-rotor UAVs. Fixed-wing UAVs can vary in wing shape and configuration, including normal (straight), swept-back, swept-forward, and delta wings [18]. They are known for their stability, larger size, and superior flying capabilities, including better cargo capacity and endurance. However, their size makes them less versatile than rotary-wing UAVs.

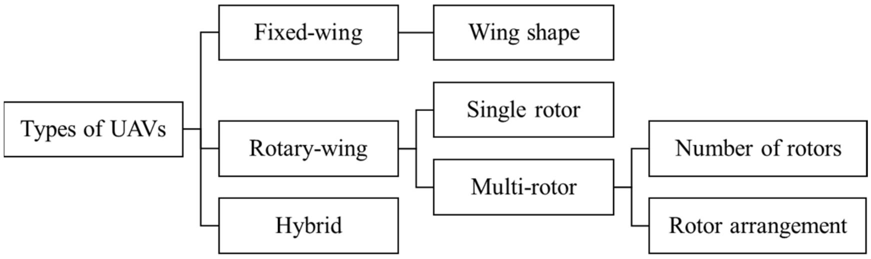

Figure 2. This diagram illustrates the three main categories of UAVs.

On the other hand, rotary-wing UAVs, or rotorcraft, generate lift with a single rotor or multi-rotor (two rotors or more). Multi-rotor UAVs are categorized based on the number of propellers into quadcopters, hexacopters, and octocopters. Additionally, multi-rotor UAVs can have variations in rotor arrangements, such as H, I, X, or Y configurations, which impact their flight dynamics and stability. These rotors, attached to the center body and pointed upwards, enable these UAVs to maintain a constant altitude and utilize their vertical take-off and landing (VTOL) capabilities. Despite their smaller size, rotary-wing UAVs offer more excellent maneuverability and versatility than fixed-wing UAVs. Additional efforts have been made to develop hybrid UAVs that combine the gliding capabilities of fixed-wing UAVs with the vertical takeoff and landing (VTOL) capabilities of rotary-wing UAVs [17,18].

When selecting the most suitable UAV type for a specific ITS application, several factors must be considered, including the UAV's physical requirements, such as size, weight, maneuverability, flight time, and energy capacity. The size of the UAV should be appropriate for the type of operation as it directly impacts the total flight time and overall efficiency. For example, smaller and lightweight UAVs (typically under 2 kg) are primarily used for recreational purposes. Slightly larger UAVs (ranging from 2 kg to 150 kg) are commonly employed by private companies for tasks such as delivering packages, aerial photography, crop dusting, or powerline/pipeline inspection. Fixed-wing UAVs (exceeding 25 kg) are predominantly utilized by the military for armed reconnaissance and surveillance operations [14]. Rapid maneuverability is another critical attribute to consider in the design, as UAVs could encounter unexpected obstacles and sudden weather changes. Also, the weight of the UAV needs to be considered in the design since the required energy level and the UAV's physical load are related. The UAV's physical load must be as light as possible to maintain the necessary energy level. A heavier UAV structure requires more energy to maneuver, often resulting in a larger battery pack. The weight of a UAV is closely tied to the challenges of payload transportation. Several advancements, such as energy-based control (EBC), can stabilize the vehicle's translational dynamics and reduce payload oscillation. Additionally, a nonlinear state feedback controller utilizing linear matrix inequalities (LMI) can effectively manage the quadrotor's rotational dynamics [19].

Selecting the appropriate battery per the energy requirement is another crucial step toward designing an efficient UAV system [20]. Lithium polymer (LiPo) batteries are often employed as energy sources in UAVs due to their ability to provide a large amount of power, considering their size [21]. However, these batteries could be over-discharged or overcharged, leading to ignition. The voltage range for LiPo cells is approximately 3 V to 4.5 V. The type of motors used by UAVs often need higher and constant voltage to function properly. As an alternative to LiPo, Lithium-ion (Li-Ion) batteries offer higher energy density, making them preferable in the application of larger UAVs that require longer flight times [22]. Another alternative is lithium iron phosphate ($LiFePO_4$) batteries, known for their high thermal stability and extended cycle life. $LiFePO_4$ batteries are often preferred in applications where safety and longevity are prioritized, although they are typically heavier and have lower energy density than LiPo and Li-Ion [23].

Moreover, UAV software helps in flight planning, stores and processes sensor data, and provides intelligent navigation. Selecting the appropriate UAV software is crucial, depending on the specific application. The minimum requirements for effective UAV software should include automated flight plans, augmented views, and image geo-rectification [24]. Some software currently used for data capturing and processing in UAV applications includes Pix4Dmatic, DJI Terra, Global Mapper, Trendspek, and Nira [25]. Software verification also plays a vital role in safeguarding UAVs from cyber-attacks, ensuring the integrity and reliability of the data collected, preventing unauthorized access to control systems, and maintaining overall operational security.

2.1.2. Architectural View

Utilizing UAVs in an intelligent traffic management system can offer substantial benefits, including broader area coverage, greater flexibility, and improved cost-efficiency compared to traditional ground-mounted solutions [13]. However, several factors must be considered when deploying UAVs optimally for successful traffic monitoring and data acquisition. This subsection will discuss recent developments in modeling and solving UAV deployment challenges.

In UAV-enabled ITSs, the architectural view of deployment strategies and trajectory optimization are closely interrelated [26]. Effective deployment strategies rely on optimized trajectories to achieve mission objectives efficiently. Deployment strategies involve the broader planning of where and how to position UAVs for various missions, considering logistical and operational considerations. In contrast, trajectory optimization focuses on refining the specific flight paths of UAVs to enhance efficiency, safety, and task performance. Table 1 summarizes recent research progress on UAV deployment and trajectory optimization.

Table 1. Research summary on UAV deployment and trajectory optimization.

References	Research Area	Methodology	Findings
Koyuncu et al. [26]	Deployment and trajectory optimization of multiple UAVs	Quantization theory	The optimal UAV locations are found to minimize the average power consumption or maximize the data rate of GTs.
Guerrero-Sánchez et al. [27]	Quadrotor trajectory tracking	Filtered observer-based IDA-PBC strategy	Able to guarantee a near-zero steady-state error in situational uncertainties and disturbances.
Dai et al. [28]	Multi-objective deployment optimization	Reinforcement learning-based approach	The number of UAVs is minimized for an efficient deployment scheme.
Brust and Strimbu [29]	UAV deployment and network topology	Networked swarm model	The proposed swarm movement is as fast as single UAVs for reaching the destination location.
Zhang and Duan [30]	Emergency UAV deployment	Optimal deployment algorithm to balance UAVs' diverse flying speeds and coverage	The proposed algorithm computation complexity of $O(n^2)$ is relatively low.
Ruan et al. [31]	Energy-efficient UAV deployment	Deployment model for multi-UAV coverage	The UAV coverage problem is solved with an energy shortage scenario.
Zhao et al. [32]	Deployment for demand coverage	Centralized deployment algorithm and distributed motion control algorithm	Enables UAVs to control motion and on-demand coverage autonomously.
Wang et al. [33]	UAV deployment and communication	Adaptive deployment scheme	The algorithm helps UAVs adapt displacement direction and distance.
Liu et al. [34]	Deployment and movement	Experience-driven dynamic movement and deployment	The proposed algorithm shows faster convergence.
Fu et al. [35]	Deployment for IoT	Both deployment and transmission power of UAVs are optimized	The deployment problem was solved to control the power of the IoT.
Valiulahi and Masouros [36]	UAV deployment	3D multi-UAV deployment approach	The minimum achievable system throughput is maximized.
Huang et al. [37]	UAV network deployment	A decentralized navigation scheme	Enable the UAV to have better views of the ground vehicles by finding the blockage area.

Note: GTs = Ground Terminals, IDA-PBC = Interconnection and Damping Assignment-Passivity Based Control, 3D = Three Dimensional, IoT = Internet of Things.

UAV-enabled networks are often quite dynamic due to the ever-changing location of UAV nodes. The dynamicity of UAV networks offers many advantages while aiding the ITS, including improved coverage, the ability to keep pace with the on-road vehicles for monitoring and tracking, and gathering traffic information. However, it also comes with several theoretical and functional challenges. Common UAV deployment problems include finding optimal UAV locations, optimizing UAV trajectories, deployment optimization for speed, coverage, and energy efficiency, emergency response, network topology, communication for the Internet of Things (IoT), and data acquisition. Several optimization algorithms have been developed lately for static and dynamic deployment to address those challenges. Alzenad et al. [38] addressed the static deployment problem, considering the optimal placement of UAVs for maximum coverage. Some works considered other constraints, including average throughput [39] or speed, time, and communication [40].

Moreover, dynamic deployment has also been studied in different scenarios. The dynamic deployment problem must be considered for UAV-enabled ITSs due to the ever-changing variation in the coverage area, on-road vehicle location, speed, time, and on-time responsive communications. Koyuncu et al. [26] employed quantization theory to optimize UAV deployment and trajectories across multiple ground terminal (GT) scenarios. They developed models for static and dynamic cases: the static case aimed to minimize average GT transmission power, while the dynamic case focused on reducing time-averaged power consumption. The deployment of UAV-mounted base stations has become an emerging solution for communicating with GTs. Dai et al. [28] modeled the deployment problem of UAVs considering the constraints on transmitting power, number of UAVs, caching, and locations. The proposed model aims to provide the required quality of communication services to GTs while maximizing the benefit of caching.

In ITS application, one of the main goals of the UAV deployment optimization problem is establishing an efficient traffic monitoring scheme. While a single UAV with a fixed trajectory could provide coverage of a specific region of the transportation networks, the use of UAV swarms would be more adaptive to trajectories and beneficial for improved detection and tracking performance. Figure 3 shows an example of the UAV swarm structure during different flight phases. UAV swarm architecture involves coordinated communication for mission planning (pre-flight), real-time data exchange and adaptive behavior (during flight), and comprehensive data analysis and system diagnostics (post-flight).

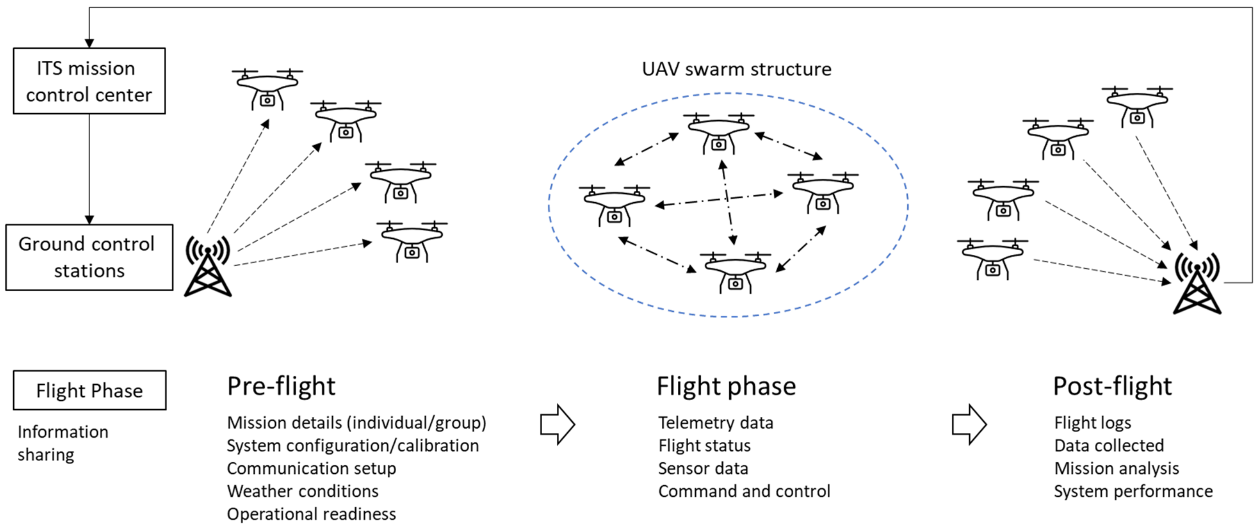

Figure 3. UAV swarm architecture during different flight phases.

The implementation of UAV swarms and the associated challenges have been investigated for various purposes. Kyrkou et al. [41] explored an optimized vision-directed deployment strategy for rapid traffic monitoring. The optimization problem is formulated to find the lowest number of UAV hovering locations subject to the data acquisition, avoiding building heights caused by Field-of-View (FoV) obstructions. The proposed problem solution includes a deployment map that can capture road traffic footage to help assess congestion levels. Kuru [42] explores the optimization of swarms of fully automated UAVs (FAUAVs) to utilize the benefits of autonomous UAV swarms for smart cities.

One of the most significant challenges of deploying UAV swarms is maintaining continuous inter-UAV communications. Brust and Strimbu [29] developed a networked swarm model to manage UAV movements in complex environments. Their approach addresses the challenge of inter-communication among UAVs amid obstacles like trees and infrastructure. For high-quality forest mapping, they employed a leader election algorithm that collects data from the swarm and directs it to the target location. Moreover, deploying a UAV network aims to provide the optimal coverage of ground nodes. Reina et al. [43] modeled this problem as a multi-objective coverage problem considered a nondeterministic polynomial time (NP)-hard problem. A multi-layout multi-subpopulation genetic algorithm (MLMPGA) was used to solve the problem. Another major challenge of deploying UAV swarms is collision avoidance. Fabra et al. [44] identified and reviewed some technical challenges and current developments associated with collision-free coordination and interaction between UAVs.

2.2. Second Layer: Communication Networks for Traffic Monitoring

The second layer of the UAV-enabled ITS application focuses on using advanced communication technologies to ensure that data from sensors and UAVs are transmitted and processed in real time. This aims to improve traffic management by enhancing connectivity, reducing latency, and enabling timely responses to traffic conditions.

2.2.1. Communication Networks

Unmanned aerial vehicle communication networks (UAVCNs) are composed of multiple UAVs working together in ad hoc mode while being controlled from a ground terminal to carry out a mission, as shown in Figure 3. The UAVCNs require effective coordination and collaboration to perform the desired mission, which involves complex networking models and mechanisms. UAVCNs are a type of Flying Ad Hoc Network (FANET), which is related to Mobile Ad Hoc Networks (MANETs) and Vehicular Ad Hoc Networks (VANETs) [45]. Additionally, UAVs integrated with vehicles support vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2X) communications in ITSs.

UAVCN system architectures characterize UAV networks based on several factors, including the network architecture (such as mesh topology and hierarchical structure), node roles (such as UAV nodes, ground stations, and gateway nodes), and communication protocols (such as routing protocols and data link layer protocols), as shown in Figure 4 [46].

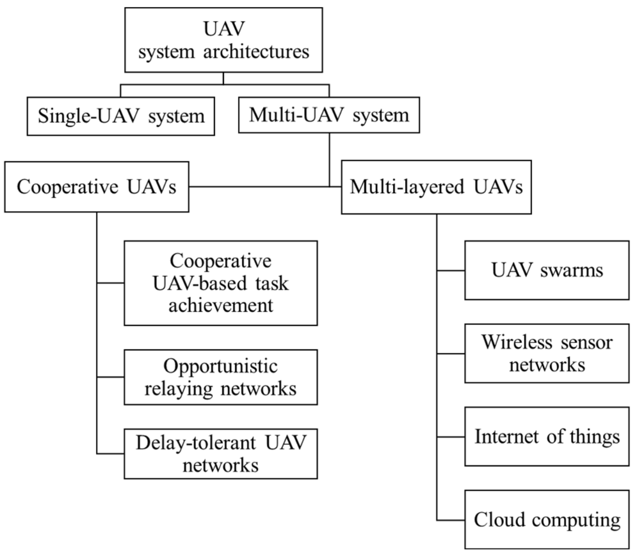

Figure 4. Categorization of UAV system architectures [46].

According to the general requirements, the UAV network could consist of one or more large or small UAVs and other connecting components. The large UAVs could be used individually in a single network, while the small UAVs could form swarms or multi-UAV systems. As shown in Figure 5, while single-UAV systems have a relatively simple communication network, multi-UAV systems introduce significant complexity due to the need for inter-UAV communication, advanced protocols, scalability concerns, and secure network management. In multi-UAV systems, multiple UAVs are interconnected with ground nodes (or stations) to form a coordinated two-way communication network. These systems rely on two main types of communication: UAV-to-infrastructure, where drones communicate with ground stations or base stations, and UAV-to-UAV, where drones exchange information directly with each other. The complexity of UAVCNs in multi-UAV systems increases with the number of UAVs and the level of intricacy of the mission.

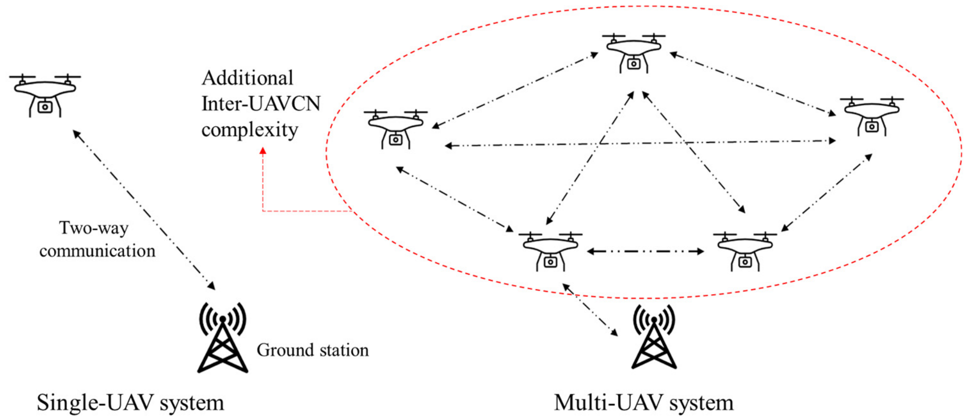

Figure 5. Complexity of communication network between single-UAV and multi-UAV systems.

Multi-UAV systems can be categorized into two primary types as follows:

1. Cooperative Multi-UAVs: In these systems, UAVs collaborate, sharing information and tasks to achieve common goals more effectively. This approach is beneficial for missions that involve multiple UAVs, each potentially having different communication capabilities. Examples of cooperative multi-UAV architectures include cooperative task achievement, opportunistic relaying networks, or delay-tolerant networks.
2. Multi-Layered UAV Systems: These systems are structured into different hierarchical layers, with UAVs assigned specific roles and responsibilities at each layer. This layered approach helps manage complex missions more effectively by organizing tasks and communication. In addition to interacting with each other in UAV swarms, UAVs in multi-layered systems often interact with IoT devices for enhanced data collection and processing, ground wireless sensor networks (WSNs) to gather and relay environmental data, and cloud computing for data storage, processing, and analysis, allowing for more scalable mission support.

Multi-UAV systems often provide significant benefits over single-UAV systems. Although multi-UAV systems can be costly, they offer more benefits due to the increased efficiency and enhanced coverage in ITS applications. Table 2 compares single-UAV and multi-UAV systems based on scalability, speed, bandwidth, antenna, control complexity, cost, survivability, and failure.

Table 2. The difference in features of single-UAV and multi-UAV systems [47].

Features	Single-UAV System	Multi-UAV System
Scalability	Limited	High
Speed	Slow	Fast
Bandwidth Required	High	Medium
Antenna	Omni-directional	Directional

Table 2. *Cont.*

Features	Single-UAV System	Multi-UAV System
Control Complexity	Low	High
Cost	Medium	Low
Survivability	Poor	High
Failure Impact	High	Low

UAVCN design considerations are often dynamic and require a higher level of coordination, making the system highly complex. The following factors must be considered while designing a UAV network to avoid unwanted interruptions during the mission.

- Mobility: UAV networks often have high node mobility and speed, which could result in several communication design challenges. The degree of mobility of UAVs would vary depending on the application. For disaster management, the UAVs would fly over the designated area and provide communication at a slow or dynamic speed. On the other hand, the UAVs could cover a large area with more dynamic links while used for surveillance.
- Adaptability: UAV network parameters can be changed considering the applications and circumstances. UAV networks should be able to adapt to common changes, such as the locations of the UAVs, the distance between UAVs, and the routes. Moreover, UAV networks are often affected by unexpected weather changes. In most applications, UAV networks need to adjust to any of these changes.
- Reliability: UAV networks are vulnerable to various natural phenomena or human attack, which can diminish performance. To withstand such attacks, UAV networks need to be robust and reliable, and reliability depends on the application-specific parameters. Comprehensive analyses, delay-tolerant functionalities, and recovery algorithms are required to ensure high reliability in the network. Moreover, the degree of reliability of UAV networks could also vary depending on the task.
- Scalability: UAV networks should be highly scalable to allow the addition or subtraction of UAVs with minimal restoration and reorganization costs. The scalability of a UAV network depends on both the nodes and the resources. The scalability of UAV networks could be increased by hiding any obstacles around the coverage area and adding more relay nodes.
- Bandwidth requirements: The bandwidth of a communication system should align with the transfer rate requirements of the UAV's specific application. ITS applications that involve high-resolution video streaming, real-time data transfer, or complex sensor data generally require higher bandwidth to function effectively.
- Power consumption or Energy considerations: UAV network power depends on the UAV's energy source. Most UAVs are battery-operated with a limited energy source. A mission's power consumption significantly impacts the network's lifespan. Therefore, designing an energy-efficient network configuration is essential to ensure longevity and optimal performance.
- Coverage: The design of a UAV system must ensure comprehensive area and network coverage within budget and time limits. This includes accounting for the UAVs' movement patterns to avoid gaps in monitoring and maintain reliable communication. An adequate coverage design balances operational needs with constraints to optimize performance.
- Security and Privacy: UAV networks can be viewed as cyber-physical systems (CPSs) because they integrate physical components (such as sensors and actuators) with digital systems (such as communication equipment and the internet). Securing the data and providing the user with the required privacy is challenging. The variety of components adds to the network's complexity, making it more vulnerable to potential security issues.

2.2.2. Advances in Traffic Monitoring

Proper monitoring of the transportation system is a prerequisite for the proactive management and effective control of ITSs. In UAV-based traffic monitoring, a level of human interaction is involved in collecting traffic flow parameters and conducting further analysis to assess traffic behavior and the performance of road geometries. Different traffic behaviors (congestion, shock waves, lane changes, gap acceptance, and accidents) and road geometries (roundabouts and signalized and non-signalized intersections) are monitored using UAVs. A summary of the research on traffic monitoring using UAVs can be found in Table 3.

Table 3. Research summary on advances in traffic monitoring using UAVs.

References	Research Area	Methodology	Findings
Khan et al. [48]	UAV-aided traffic monitoring systems	A systematic review of UAV-based traffic monitoring studies	A universal multistep traffic monitoring framework.
Barmpounakis et al. [49]	UAV-aided traffic monitoring literature	Review of studies and applications incorporating unmanned aircraft systems (UASs) in transportation	Reveals current issues and emerging challenges.
Barmpounakis and Geroliminis [50]	Big data-based traffic congestion analysis	Investigating traffic congestion by creating an extensive data repository of UAV-based traffic stream data	Offers an opportunity to test traffic models developed from diverse disciplines.
Elloumi et al. [12]	Adaptive multiple UAV-aided real-time traffic monitoring system	Adaptive trajectory-based multi-UAV traffic monitoring	Better coverage rates and event detection rates.
Niu et al. [51]	General UAV-based traffic monitoring with cloud processing	Cloud processing and UAV-based traffic monitoring	Monitors traffic with high accuracy and flexibility.
Li et al. [52]	Traffic monitoring based on UAV scheduling problem	Scheduling of UAV routes under uncertain demands	Competitive performance from a local branching-based solution method.
Khan et al. [6]	UAV-based smart traffic monitoring	5G technology and UAV-based traffic monitoring	Better projection of the number of accidents.
de Frias [53]	Cooperative traffic monitoring for drivers	UAVs collect and share vehicle data with other vehicles	Cooperation is an effective tool for traffic monitoring.
Choi and Lee [54]	Rapid monitoring with a high-quality digital elevation model	UAV-based rapid aerial mapping system	The rapid traffic monitoring system successfully passes the integration test.
Doherty et al. [55]	Developing technologies for autonomous UAVs operating over large-scale transportation networks	Multidisciplinary basic research approach to developing autonomous UAVs	Requires multidisciplinary research competencies.
Nordberg et al. [56]	Wallenberg Laboratory for Information Technology and Autonomous Systems (WITAS) UAV project	Customization and incorporation of vision algorithms to meet the needs and requirements of the UAV platform	Presents research work by the computer vision laboratory at Linköping University (LiU).
Rasmussen et al. [57]	Autonomous UAV-based system to track traffic on a road network	Patrol, isolation, capture, and delivery—four-phase approach to monitor and detect intruder vehicles by cooperatively using UAVs and unattended ground sensors (UGSs)	Demonstrates autonomous decision-making of cooperative UAVs and UGS.
Rasmussen et al. [58]	Development of an open-source software framework named UxAS	A software framework for autonomous UAV-based systems	Facilitates the development of new autonomous UAV-based systems.

With technological advancements, most research work includes using the latest wireless technologies to ensure uninterrupted communications between UAVs and GTs. Khan et al. [48] studied various UAV-aided traffic monitoring systems that many researchers have proposed over the years with a detailed step-by-step framework. A review of the literature on UAV-aided traffic monitoring is presented in Ref. [49]. An extensive data-based system for traffic congestion analysis in large-scale transportation networks is proposed by Barmpounakis and Geroliminis [50]. Using a swarm of 10 drones, the authors recorded traffic streams from an urban area for five days and analyzed the data to extract traffic behaviors at both microscopic and macroscopic levels.

Elloumi et al. [12] proposed a multiple UAV-aided real-time traffic monitoring system. In this system, UAVs adaptively adjust their trajectories, collect vehicle information from city roads, and send them to a data processing center to help make traffic management decisions. The proposed system performed better than the traditional fixed trajectory UAV-based traffic monitoring systems, considering the metrics of vehicle coverage and event detection rates.

Short-term traffic monitoring using static solution methods is limited by a lack of flexibility, expensive sensors, and low coverage. To overcome these limitations, Niu et al. [51] presented a general UAV-based traffic monitoring system using an onboard camera to record traffic flow information. The UAV-collected video data were then uploaded to a cloud server, where vehicles were detected from the video data using the Haar cascade algorithm and tracked using a frame-by-frame comparison algorithm. The experimental results demonstrated that the proposed traffic monitoring system is highly accurate and flexible. Li et al. [52] investigated real-time traffic monitoring systems from the perspective of UAV scheduling problems with uncertain demands. The authors designed a mixed-integer linear programming (MILP) model for the scheduling problem and proposed a local branching-based solution method. The suggested approach could quickly find a better solution while still approximating the actual solution. A Shanghai road traffic case study demonstrated the MILP model's effectiveness and use of the solution method.

Khan et al. [6] proposed a UAV-based smart traffic monitoring system to overcome limitations and loopholes. The smart system consists of three layers as follows: the first layer is the monitoring layer, where a UAV equipped with an advanced camera is deployed to monitor highway traffic and detect speed and other traffic violations; the second layer consists of a 5G communication technology, which connects the first layer with the base station; and the third layer consists of highway traffic where the actual violations take place. The proposed intelligent traffic monitoring system helped reduce accidents and fatalities in the projected results.

A cooperative traffic monitoring system was proposed to help drivers with surrounding vehicle information by de Frías et al. [53]. In this system, traffic images are taken using UAVs and sent as input to a semantic neural network. The neural network provides a segmented vehicle image as output with an attached 2D geographic location. These output data are then sent to other vehicles to inform them about the location of the vehicle on the road. Several experiments have demonstrated that the proposed traffic monitoring system is efficient and robust. Choi and Lee [56] introduced a rapid monitoring system based on UAVs to facilitate emergency response in real time. An integration test was conducted to evaluate the performance of the proposed system. While performing tests, data were acquired from the test site in real time, and by processing the data, a high-quality digital elevation model and orthoimages were generated. However, the usage of UAVs in actual road traffic management is still under investigation [48].

Much research has been conducted to develop autonomous UAV-based traffic monitoring systems. Figure 6 shows an example of a general traffic monitoring scenario using UAVs in coordination with service stations and roadside units (RSUs). Doherty et al. [55] originally proposed the Wallenberg Laboratory for Information Technology and Autonomous Systems (WITAS) UAV project, which aimed to successfully develop methods and technologies necessary to operate autonomous UAVs over large-scale transportation networks. The project was multidisciplinary and required many different research competencies. The UAV was designed to have automatic navigation capabilities at various altitudes, including take-off and landing. The UAV was expected to perform multiple vehicle detections, tracking, and parameter extractions to achieve a specific mission goal and to identify complex traffic behaviors, such as vehicle overtaking, traversing of intersections, and parking lot activities.

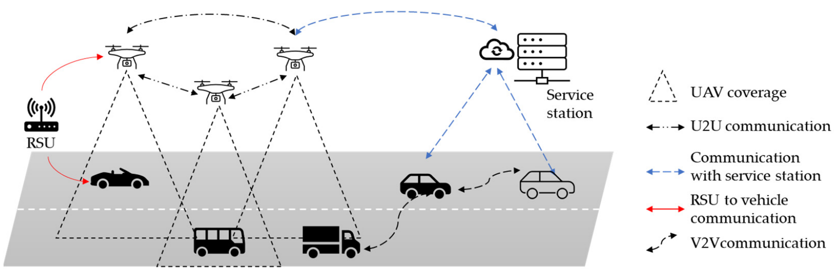

Figure 6. Traffic monitoring scenario using UAVs.

Parameter perceptions from operating environments and devising and implementing plans based on these perceptions are essential abilities of an autonomous UAV system. Although many research works have built and implemented plans in their autonomous systems, few efforts have been made to perceive the environmental parameters and devise a plan accordingly. Rasmussen et al. [57] proposed an autonomous system consisting of ground sensors and UAVs to track traffic on a road network. In their proposed system, each UAV can communicate with unattended ground sensors and other UAVs to find traffic of interest. The authors demonstrated autonomous decision-making by conducting field experiments where two UAVs autonomously collaborated to track a moving ground vehicle.

To facilitate the development of autonomous UAV systems, Rasmussen et al. [58] introduced an open-source software framework named UxAS for prospective teams of unmanned systems. UxAS offers several cutting-edge features, such as dynamic inter-UAV communication, basic autonomous capabilities, support for incorporating new autonomous capabilities, and manageable software complexity with service-oriented architecture. This open-source framework brought out an opportunity to develop new autonomous UAV systems for responsive traffic monitoring. The authors compared UxAS and robotics software frameworks (RSFs), described the core services of UxAS, and discussed their working procedures for executing multi-vehicle missions.

2.3. Third Layer: Data Collection and Processing

The third layer focuses on data collection and processing, involving aggregating and analyzing information from various sources, including sensors and UAVs. This layer transforms the raw collected data from UAVs into actionable insights for better decision-making and improved system efficiency.

2.3.1. Data Collection

In ITSs, traffic data are collected, processed, and used for predictions that enable proactive management of the transportation system. Table 4 presents a brief research summary of advancement in UAV-assisted data collection and post-processing methods.

Table 4. Research summary of data collection and post-processing methods.

References	Research Area	Methodology *	Findings
Du et al. [59]	Traffic data	Provides algorithms for internal and external camera parameters calibration and for fixing image distortion	The data platform helps verify traffic simulation models
Ke et al. [60]	Traffic parameter estimation	Three functional modules for estimating microscopic and lane-level macroscopic traffic parameters	The framework successfully extracts microscopic and macroscopic traffic parameters
Ke et al. [61]	Traffic parameter estimation	Ensemble classifier and optical flow-based four-step framework for estimating real-time traffic flow parameters	The framework can effectively estimate traffic parameters in congested traffic conditions
Ke et al. [62]	Traffic parameter estimation	Kanade–Lucas–Tomasi (KLT) tracker, k-means clustering, connected graphs, and four-step traffic flow framework	The framework successfully estimates bidirectional traffic parameters
Brkić et al. [63]	Traffic parameter estimation	A traffic parameter estimation framework using terrain survey, image processing, and vehicle detection methods	Enables extraction of trajectories for each individual vehicle for microscopic traffic flow estimation
G Leduc [64]	Traffic data collection methods	A short overview of different traffic data collection methods and traffic parameter estimation methodologies	Floating car data (FCD) collection methods are cost-effective and provide better coverage than traditional methods.
Zhu et al. [65]	Traffic parameter estimation	A deep learning-based vehicle counting framework	The framework performs better than other deep neural network-based techniques
Khan et al. [66]	Traffic trajectory extraction	An automated UAV video processing framework to extract vehicle trajectories	The framework is supported by a field experiment
Khan et al. [67]	Analysis of traffic information	A case study of an automated vehicle trajectory extraction method in shockwave analysis at signalized intersections	Depicts the applicability of the UAV-based traffic analysis system
Van Lint et al. [68]	Traffic and travel time prediction methods	A taxonomy of short-term traffic prediction methods	None of the prediction models perform best in every possible traffic situation
Wu et al. [69]	Traffic prediction under abnormal conditions	An online boosting nonparametric regression (OBNR) model for traffic forecasting	The OBNR model performs better than traditional online learning models in abnormal traffic conditions
Shepelev et al. [70]	Traffic flow prediction in intersections	Multiple regression analysis-based traffic characteristics identification and traffic throughput prediction by fuzzy logic	Reduction in pedestrians' waiting time by up to 30%

Table 4. *Cont.*

References	Research Area	Methodology *	Findings
Zhao et al. [71]	Short-term traffic forecast	A long short-term memory (LSTM)-based traffic forecast model	Provides robust traffic volume forecast
Afrin and Yodo [72]	Traffic flow prediction	A framework for correlated traffic prediction using an LSTM network	Performs better than the LSTM and SARIMA models
Koesdwiady et al. [73]	Traffic flow prediction	A deep learning and data fusion architecture for traffic prediction	Provides more accurate traffic prediction
Guo et al. [74]	Stochastic traffic flow prediction	An adaptive Kalman filter approach for processing SARIMA + GARCH structure	Demonstrates improved adaptability in highly volatile traffic situations

* Notes: SARIMA = Seasonal autoregressive integrated moving average, GARCH = Generalized autoregressive conditional heteroscedasticity.

Traffic data collection methods can be divided into two broad categories: conventional "in situ" technologies and floating car data (FCD) technologies [64]. In conventional "in situ" technologies, traffic data are collected using sensors located along the roadside. The data collection methods of this category can be further categorized as intrusive and non-intrusive methods. Intrusive methods are direct methods where a sensor and a data collector are placed on or in the road. Different data collection methods, such as induction loops, pneumatic road tubes, piezoelectric sensors, and magnetic loops, fall into this category. In non-intrusive methods, traffic data are collected based on remote observations. Many data collection methods, such as manual counts, passive and active infrared, ultrasonic and passive acoustic, and video image detection, fall into the non-intrusive category. Even though the "in situ" technologies have matured and provide high-quality traffic data, they suffer from high installation and maintenance costs, limited coverage of the transportation network, and vulnerability to bad weather conditions.

In the FCD technologies, real-time traffic data are collected via mobile phones or global positioning system (GPS) information [64]. In this category, the traffic data collection sensors are drivers' mobile phones and vehicles' GPS information, and the data are collected and stored anonymously in a central processing center. One of the drawbacks of GPS-based FCD technology is limited data coverage, as the old vehicles are rarely equipped with GPS, and new installation costs are high, even though modern vehicles are equipped with GPS. On the contrary, the cellular-based FCD technology has better coverage as mobile phones are cheaper and more common than GPS. Collecting and processing real-time traffic data and subsequently notifying drivers of potential congestion, alternative routes, and incident information greatly enhances the effectiveness and efficiency of the intelligent transportation system (ITS). Still, the real-time tracking of cell phones exposes a considerable security threat for individuals.

UAVs offer an effective alternative for traffic data collection over traditional methods because of their spatial and technical advantages. UAV-enabled traffic data collection methods perform better than conventional "in situ" methods, as the UAVs are flexible and relatively inexpensive, and can cover a wide range of areas, whereas conventional "in situ" methods are fixed-positioned, expensive, and can cover only a limited area in the transportation networks. Compared with FCD technologies, UAV-enabled traffic data collection methods perform better regarding the extent of traffic data coverage and individual security concerns. When vehicles operate in the transportation network without cell phones and GPS, FCD technologies cannot collect their information. Moreover, every piece of data collected via FCD technologies can be traced back to an individual or family, which poses a security issue. On the contrary, UAV-enabled traffic data collection methods can provide complete data coverage of the transportation system without linking the data to any persons or families. Furthermore, UAV-enabled methods can extract microscopic, mesoscopic, and macroscopic traffic parameters. UAVs using digital cameras capture traffic flow data, which covers almost all traffic information, and hence, researchers working on traffic parameter extraction use video data. An example of a scenario in which traffic flow is captured on video using a UAV is shown in Figure 7. In the figure, a UAV hovering over the road uses a digital camera to take a traffic flow video.

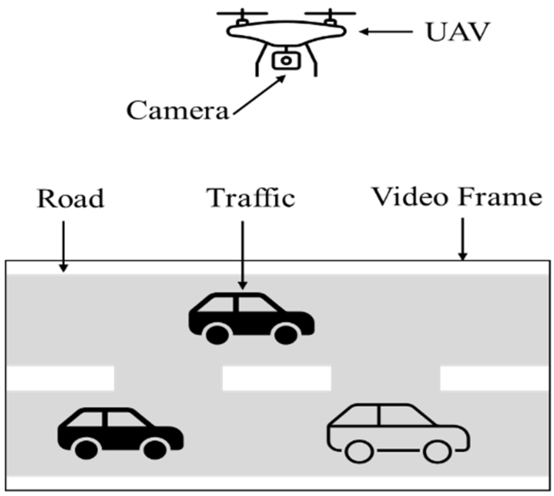

Figure 7. Traffic flow video data collection using UAV.

2.3.2. Data Processing

In a UAV-based ITS, traffic parameters are extracted from UAV-collected video data. UAV video data differ from conventional video data in terms of the background movement of the video frame. In the case of conventional video data, the background of the video frame does not change and remains fixed during the period of the video. However, in the case of UAV video data, the background of the video frame continuously changes. The movement of the background of the video frame is called ego-motion. Because of the ego-motion of the UAV video data, traditional vehicle detection and traffic parameter estimation methods are not appropriate for extracting traffic parameters from them. Research papers addressing ego-motion and extracting traffic parameters from UAV-collected data are discussed in this section.

For the accumulation of traffic data and traffic simulation model verification, Du et al. [59] proposed an open data test platform. In this platform, the authors suggested uploading highly accurate and undistorted traffic data obtained after calibrating camera parameters. The authors proposed a calibration algorithm based on the pinhole camera model to determine intrinsic and extrinsic camera parameters. The calibration of the intrinsic parameters was carried out using HALCON software [75]. Image distortion correction was performed using the model proposed by Lenz et al. [76]. The extrinsic parameters were calibrated by leveraging the orthogonal properties of rotation and translation matrices. Additionally, traffic parameters were extracted using the Scale-Invariant Feature Transform (SIFT) and Continuously Adaptive Mean Shift (CAMShift) algorithms.

Ke et al. [60] proposed a framework for assessing microscopic traffic behavior and lane-level macroscopic speed, density, and velocity traffic parameters. The framework comprises three modules: functional, data storing, and parameter estimation. The functional module extracts lane-level traffic information by motion vector processing, vehicle detecting, and vehicle tracking. The output data of the functional module are then stored in a unique data structure in the data-storing module. The traffic data stored in the data storing module are processed to estimate microscopic and macroscopic traffic parameters in the parameter estimation module. The authors proposed a new multiple vehicle tracking algorithm consisting of detection, prediction, and association steps. The algorithm uses an ensemble vehicle detector and the Kanade–Lucas–Tomasi (KLT) optical flow tracker for vehicle detection and prediction. Moreover, each predicted vehicle is associated with the corresponding detected vehicle by implementing several rules in the algorithm.

To estimate traffic flow parameters in real time for both free and congested flow conditions, Ke et al. [61] proposed a complete analysis framework for UAV video data. In this four-stage framework, vehicle detection and traffic parameter estimation are carried out in the first and last two stages, respectively. A supervised learning method is used for traffic flow parameter estimation, and an ensemble classifier is used for vehicle detection. The proposed ensemble classifier outperformed contemporary state-of-the-art vehicle detectors designed for UAV-based vehicle detection.

Ke et al. [62] proposed a four-step framework to estimate bi-directional traffic flow parameters in real time by analyzing UAV videos. In the first step, Shi–Tomasi features are used to detect interest points in each pair of consecutive frames, and the interest points are tracked using the Kanade–Lucas optical flow algorithm. In the second step, based on the speeds and directions of the interest points, the optical flow vectors for both vehicles and the background are clustered using the *k*-means algorithm. In the third step, for every vehicle, a connected graph is generated, connecting all the associated interest points. The association between a vehicle and its interest points is based on the concept that they should have similar positions and velocities. The vehicle counts in a direction can be calculated from the number of connected graphs in that direction. Traffic parameters such as speed, densities, and volumes for both the forward and reverse directions are calculated in the final step.

Brkic et al. [63] proposed a new framework for calculating traffic flow parameters. The main steps of this framework are province survey, image processing, vehicle detection, and traffic parameter calculation. For the province survey, a UAV collects traffic flow video data from a region defined by two ground control points. Image processing is carried out on the video data to narrow down the areas of interest of the video frames on the road between two ground control points. Then, in the following step for vehicle detection, a faster region-based convolutional neural network (R-CNN) is used for detecting vehicles from the truncated images, and a simple online real-time tracking (SORT) algorithm based on the Kalman filter is used for vehicle tracking. Finally, traffic flow parameters are calculated in the last step.

Zhu et al. [65] proposed a robust deep vehicle counting framework (DVCF) for counting different vehicles from UAV video data. The two main parts of the DVCF framework are deep learning-based vehicle detection and vehicle tracking. In the first part, an enhanced single-shot multi-box detector (Enhanced-SSD) was used to produce bounding boxes of vehicles containing information on the type. In the second part, a fast multi-object tracker was applied to the bounding boxes to estimate vehicle trajectory and the number of vehicles by type on the frames. The authors demonstrated that the DVCF method outperforms other existing methods in counting vehicles from high-resolution UAV video.

Khan et al. [66] provided a detailed methodological framework for automatically processing UAV video to extract multiple vehicle trajectories, which has applications in traffic parameter estimation and traffic management. The main components of the framework are preprocessing, stabilization, geo-registration, vehicle detection, tracking, and trajectory management. This framework was later used in a case study for identifying shockwaves and estimating flow parameters at signalized intersections [67]. A simple methodological framework for video data processing and traffic parameter extraction is shown in Figure 8.

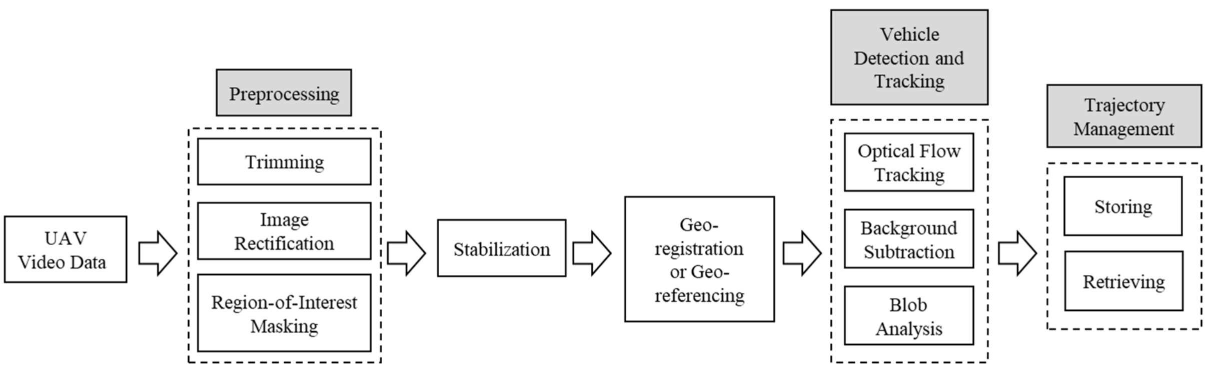

Figure 8. An automated UAV video processing and analysis framework [66].

UAV video data processing revolutionizes predictive analysis and decision-making by converting high-resolution aerial footage into actionable insights. One of the critical aspects of UAV-based ITSs is proactive decision-making based on traffic behavior prediction. Proactive decisions help better manage the transportation system in the case of predicted incidents. In a UAV-based ITS, a proactive decision support system should have the following modules: (1) UAV deployment for traffic data collection, (2) traffic data processing for noise reduction and parameter estimation, and (3) data-driven prediction of traffic behaviors. A comprehensive, proactive decision support system for UAV-enabled ITSs is still underdeveloped. Current research often addresses the components—UAV deployment, data collection, and traffic prediction—separately, without integrating all three into a unified framework. A brief description of the traffic behavior prediction techniques is presented in this section.

The categories for traffic flow prediction approaches are (1) the parametric approach, (2) the nonparametric approach, and (3) the hybrid approach [68]. In the parametric approach, the structures of the prediction models are defined in advance, and the parameters of the models are obtained from the historical traffic flow data. In the nonparametric case, the prediction model structure and the model parameters are derived from the data. The hybrid approach falls at the intersection of the parametric and nonparametric prediction methods and synthesizes the benefits of both approaches. A taxonomy of the traffic flow and travel-time prediction models is shown in Figure 9.

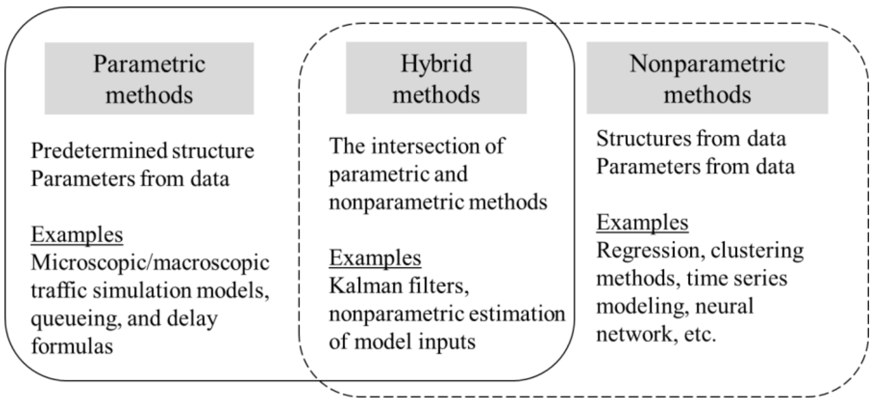

Figure 9. Taxonomy of traffic flow and travel-time prediction models [68].

Shepelev et al. [70] used multiple regression analysis to identify the major traffic characteristics responsible for the throughput of traffic intersections. The traffic data were collected using cameras, and the traffic information was extracted from the collected data using an R-CNN. After identifying the key characteristics, a fuzzy logic model was developed to predict the throughput of traffic intersections. The authors used statistical software (Statistical Package for Social Sciences (SPSS)) and the Fuzzy TECH 5.81d Professional Edition computer program for multiple regression analysis and fuzzy model development, respectively. In addition, the authors mentioned that by obtaining traffic characteristics of a particular intersection, the waiting time for pedestrians could be reduced significantly.

Forecasting traffic parameters in advance and managing traffic networks accordingly are some of the fundamental goals of ITSs. To forecast traffic parameters in a short time horizon, Zhao et al. [71] proposed a deep learning approach for developing a traffic forecast model. The forecast model was a long short-term memory (LSTM) network based on the Internet of Vehicles (ROVs), correlation analysis, and recurrent neural network (RNN). The proposed model differed from conventional forecast models by incorporating a two-dimensional network of memory units in the proposed LSTM network, which accounts for temporal and spatial correlations in traffic systems. The authors concluded that the traffic volume forecast of the proposed model was robust.

Over the years, many parametric and nonparametric methods have been developed for traffic prediction, but these techniques are often vulnerable when handling correlated traffic data, as they may suffer from overfitting, fail to account for complex interactions, and struggle with issues like multicollinearity. To make traffic predictions in the presence of spatial–temporal dependencies, Afrin and Yodo [72] proposed a framework for correlated traffic prediction. This framework exploited a correlated traffic data set to generate two modified data sets based on the data's temporal and spatial dependencies. Then, the temporal and spatial trends were predicted from the associated data sets using an LSTM-based framework. The final traffic prediction was made by taking the average of the spatial and temporal trend predictions. The experimental results showed that the framework would perform better than other naïve implementations of the LSTM and SARIMA methods.

Weather is a significant factor in transportation systems. Koesdwiady et al. [73] investigated the correlation between weather data and traffic parameters and proposed a complete architecture for traffic flow prediction. The prediction architecture incorporated deep belief networks (DBNs) and data fusion and predicted traffic flow in the San Francisco Bay Area using historical traffic and weather data. The performance of the DBNs was compared with state-of-the-art ARIMA and ANN models. In all cases, the DBNs outperformed the two models. One of the limitations of DBNs was that the memory requirement for the supervised training process was substantially more extensive than the memory requirement of the testing process.

Proactive traffic management systems need stochastic traffic flow prediction models in the presence of uncertainty. Guo et al. [74] proposed a Kalman filter approach for short-term stochastic traffic prediction and uncertainty quantification. Since both traffic flow predictions and their associated intervals can be derived from the combined SARIMA and GARCH structure, the authors employed adaptive Kalman filters to process this model in real time. The experimental results showed that the adaptive Kalman filter approach could provide good traffic prediction with associated intervals and was highly adaptable in volatile traffic situations.

2.4. Cybersecurity for ITS

An ITS is a complex framework comprising diverse components such as UAVs, roadside infrastructure, sensors, data storage and processing units, vehicles, and pedestrians, all collaborating to enhance the safety, efficiency, and sustainability of transportation networks [4]. Protecting sensitive data and the overall system from unauthorized third-party access is essential to ensuring accurate decision-making while safeguarding user privacy. ITS cybersecurity is critical, as vulnerabilities can directly impact human lives, mainly when affecting emergency services like medical delivery vehicles and ambulances. Implementing artificial intelligence (AI)-driven systems to detect and mitigate cyberattacks can significantly bolster the resilience of transportation infrastructure [77]. Additionally, establishing protocols for rapidly sharing cyber threat intelligence and mitigation strategies is vital. Expanding the workforce skilled in ITS cybersecurity will further reduce the likelihood and impact of cyberattacks [78]. Furthermore, developing contingency plans with the necessary resources and redundancies in high-risk components can help minimize the effects of cyberattacks and facilitate quicker recovery of ITS functionality after an incident [79].

Advances in cybersecurity for ITSs have been well discussed in Refs. [77,80–82]. These efforts summarized innovative approaches to securing the diverse components of ITSs, including the most frequent cyberattacks and their consequences [80], the implementation of advanced encryption techniques, intrusion detection systems, and robust authentication protocols [81], and potential AI-enabled solutions in UAV-assisted IoT applications [82]. These advances also address the importance of collaboration among stakeholders—such as government agencies, industry leaders, and researchers—to develop comprehensive strategies for mitigating cyber threats [81].

3. Current Research and Applications

The Department of Transportation (DoT) identifies six research areas in the ITS Strategic Plan 2020–2025 [83]. One of them is emerging and enabling technologies, which focuses on innovative technologies that facilitate ITS advancements. This section will discuss the current state-of-art of UAV research and applications in the transportation sector that complement the DoT ITS Strategic Plan.

3.1. Road Safety

One of the widely explored applications of UAVs in the transportation domain is road safety. The road safety-based applications of UAVs include accident investigation, risk assessment, and monitoring of the overall road network. To investigate a specific traffic incident, images collected by UAVs have been used to generate an incident scene. With the advancements in UAV technology, the current research focuses on the software capabilities related to image processing and scene reconstruction. Liu et al. [84] proposed a Structure-from-Motion (SFM) algorithm that uses UAV photogrammetry to reconstruct a 3D traffic accident scene. Skort [85] used 3D and orthophoto computer models for the same purpose. Moreover, research efforts have been made for UAV flight planning and control [86], capturing accident-related video data by varying heights and angles, data transferring methods, generating 3D models, and accuracy analysis [87].

3.2. Highway Infrastructure Management

The application of UAV technology in transportation involves managing highway infrastructures. The two most extensively investigated areas are bridge inspection and pavement distress recognition. Seo et al. [88] performed a bridge inspection using a DJI Phantom 4 drone and identified different structural damage. Lei et al. [89] proposed an algorithm to classify different types of fractures in the preprocessing images to characterize a cracked area. Wu et al. [90] used UAV images and deep learning models to assess bridge and pavement conditions. UAVs are also used in pavement distress recognition and road distress monitoring. Leonardi et al. [91] used 3D mapping data from UAVs to develop an automated method for road condition assessment. Lee [92] used the DJI Mavic 2 Pro drone to obtain UAV videos, which were then used to detect potholes on the road.

3.3. Emergency Response and Disaster Management

Unexpected events can significantly damage and disrupt the transportation system. For example, road accidents and traffic rule violations can lead to traffic congestion. Restoring normal traffic flow after such events can be challenging, as the transportation system requires a quick response. Incorporating autonomous UAVs into the system could reduce emergency response times and help mitigate potential traffic congestion caused by disasters. Zhao et al. [93] developed a framework for a UAV-assisted emergency network. This framework considers both trajectory scheduling and communication network design. To address the limitations of traffic policing and emergency response systems, Beg et al. [94] proposed a solution that is intelligent, autonomous, and UAV-enabled. Most applications of UAVs in this sector could be related to hazard monitoring, disaster response, relief, and recovery [95]. UAVs can play a vital role in search and rescue operations, as they can monitor broader areas easily compared to humans [96]. Additionally, UAVs are employed to deliver supplies and recover hazardous materials, including dry food, medicines, and water. In addition, UAVs can be deployed to enable communication when communication channels are destroyed, or transportation infrastructures are damaged [97]. UAVs could also be used during medical emergencies [98].

3.4. Delivery Services and Load Transportation

Another highly promising application of UAVs is delivering packages or parcels. UAV delivery will be the next generation of delivery services due to its advantages over the conventional delivery process [99]. UAVs are not currently affected by road infrastructure and traffic conditions. In addition to parcel delivery, UAVs are also being explored for efficient load transportation, including passenger services such as air taxis. Utilizing VTOL technology, air taxis can bypass ground traffic and offer rapid, on-demand transport between key locations [100]. However, several critical challenges remain, including trajectory tracking, navigation, and the coordination of multiple UAVs in load transportation [101].

Moreover, safety, risk management, and air traffic regulations must be addressed before UAV-based passenger transportation can be fully realized. Despite these challenges, advancements in UAV technology hold great promise for revolutionizing the transportation sector, highlighting the potential benefits of UAV applications.

3.5. Policy and Regulations

Research into policy and regulation for UAVs in ITSs addresses several critical areas to ensure safe and effective integration. It includes developing comprehensive frameworks for airspace integration, allowing UAVs to operate alongside manned aircraft while establishing clear guidelines for flight paths, altitude limits, and no-fly zones [102]. Safety and security standards are paramount, focusing on collision avoidance technologies, data privacy, and cybersecurity measures [103]. Additionally, regulations are being crafted for urban air mobility, managing UAV operations in densely populated areas, and overseeing infrastructure such as vertiports. Compliance and enforcement systems are essential for monitoring UAV operations and ensuring adherence to regulations. Public engagement and international collaboration are also crucial, aiming to align global standards and address public concerns [104]. These efforts collectively aim to balance the benefits of UAV technology with the necessary safety and regulatory safeguards, promoting UAVs' seamless integration into existing transportation systems.

4. Challenges and Future Research Directions

In recent years, immense growth has been observed in the advancements in research and applications related to UAV technology. The improvements range from advances in simple single-UAV systems to the implementation of autonomous UAV swarms in various sectors. However, challenges such as ensuring reliable connectivity, data security, and regulatory compliance persist in UAV-enabled ITS applications. This section will discuss future research directions aimed at addressing these challenges.

4.1. UAV-Enabled Smart Transportation

As the growing number of on-ground smart vehicles is increasing, the integration of autonomous UAVs with smart vehicles could primarily facilitate the transportation system. Both autonomous UAVs and smart vehicles could be operated without any human intervention. Uninterrupted communication between UAVs and smart vehicles would help enable automated traffic monitoring, tracking, and traffic data acquisition. Continuous monitoring and real-time feedback would help regulate the traffic flow and reduce congestion. However, the collaborative navigation of both autonomous UAVs and smart vehicles is not fully mature enough to be deployed. Several factors could make the implementation more complex, including large infrastructure, automation of field support systems, traffic control units, and emergency rescue teams. In future studies, the automation of the total transportation systems should be considered.

4.2. Design and Data Acquisition

Implementing UAV-enabled ITSs requires several design and software specifications that must be considered beforehand. These specification requirements depend on several factors, including the coverage area, flight time, traffic variability and monitoring scheme, real-time communication, and data acquisition. Although the recent advancements in UAV technology are already admirable, there is still a lot of scope for development in UAV designs related to ITS applications. Various actuators and sensors, fault-tolerant controls, and software need to be installed for traffic monitoring and tracking, which requires both a high energy supply and expertise. Moreover, uninterrupted communication among UAV swarms, vehicles, and control units is crucial. However, sudden malfunctions in the device and software can interrupt the continuous functioning of the system. Using alternative energy resources and real-time optimization can be investigated to enable longer flying time and ensure uninterrupted traffic monitoring. UAV-enabled ITSs provide a valuable source of big data related to traffic. Currently, visual and tracking data are collected using GPS information and cameras. In the future, developing an energy-efficient, integrated image processing unit for UAVs could help reduce data processing time and energy consumption.

4.3. Privacy and Cybersecurity

Maintaining the cybersecurity and privacy of gathered data is crucial in UAV-enabled ITSs. Extensive research has been conducted on these issues. In UAV-enabled ITSs, system components, including UAVs, various vehicles (air, ground, water, and rail), and control units, are mainly interconnected through wireless networks. This connectivity increases the network's vulnerability to potential cyberattacks. In addition, the cloud contains a lot of data about vehicles and passengers, which need to be protected. These data include vehicle identity, location, and traffic data, which can be hacked and used maliciously. Moreover, failing to protect these data could compromise user privacy. As ITS technology advances, cybersecurity challenges will increase, putting the system and users at greater risk without robust privacy and security protocols. Potential initiatives to address these challenges include developing intrusion detection systems, improving UAV communications, optimizing security and performance through data encryption, and applying advanced technologies.

4.4. Energy-Efficiency and Reliability

Both UAVs and smart electric vehicles are battery-operated. Many conventional UAVs typically have a battery life of less than half an hour, which limits their operational time. Since ITS operations can be demanding, requiring longer flight times, multiple sensors, and continuous network connectivity, conventional UAVs may struggle with insufficient battery life [20]. Advancements in battery materials and design are crucial to address these limitations, and exploring the integration of solar and other renewable energy sources could be beneficial. Additionally, optimizing energy consumption through enhanced flight time, operability, and coverage should be investigated to improve the performance of UAVs in ITS applications [31].

4.5. Quality 5.0

Integrating Quality 5.0 into UAV-based transportation systems presents promising avenues for future research, particularly in making these systems more human-centric and enhancing cybersecurity [105]. One key area is the development of AI-driven algorithms for optimizing UAV flight routes, communication networks, and predictive maintenance to improve efficiency and system reliability [77]. Additionally, more work is needed to develop real-time data analytics frameworks that can process large volumes of traffic and environmental data, enabling faster incident detection and response [106]. Another important direction is the exploration of UAVs as communication relays in areas with limited or disrupted connectivity, ensuring consistent network performance [40]. Lastly, human-centric interaction models, where humans supervise UAV operations through intelligent dashboards, require further investigation to ensure seamless human–machine collaboration in future intelligent transportation networks [107].

4.6. Experimental Implementations

UAV-enabled ITSs represents the future of transportation systems and remain an emerging research domain. Despite significant advancements, real-life implementation is still under investigation. The lack of comprehensive information, the internal and external uncertainties, and the absence of extensive protocols make it challenging to identify the specific obstacles to UAV integration in future ITSs. To uncover potential risks, future research should focus on experimental implementations of UAV applications within ITSs. Conducting thorough research and experiments before full-scale implementation will pave the way for successful deployment and expand the possibilities for UAV-enabled ITSs.

In addition to the potential research areas discussed in this section, numerous other avenues exist for enhancing the large-scale implementation of UAV-based ITSs. As technology continues to evolve, addressing these additional areas could significantly advance the capabilities and effectiveness of UAV-based ITSs.

5. Conclusions

UAV-enabled ITSs will be among the most radical advancements in future transportation systems. This paper provides an overview of the latest developments in various sectors related to the use of UAVs for ITSs, organized within a three-layered framework.

The first layer of the UAV-enabled ITS framework focuses on the technical specifications of UAVs, which are crucial for effective integration into transportation systems. The design and specifications of UAVs must consider factors such as size, weight, maneuverability, flight time, and energy capacity. These factors influence the suitability of UAVs for specific ITS applications. Moreover, effective UAV deployment and trajectory optimization are crucial for maximizing their benefits in ITSs. This layer explores research on methods for optimizing UAV locations, trajectories, and communication networks to ensure efficient and reliable operation.

The second layer of the framework comprises the communication networks between UAVs and vehicles and the responsive traffic monitoring system. Different types of UAV communication networks (UAVCNs) are discussed, and some network design considerations are identified. Additionally, the recent advancements, applications, and challenges in implementing UAVCNs are identified. The UAV-assisted vehicle ad hoc network communication architecture can provide significant coverage for traffic monitoring and efficient emergency services. However, implementing UAVCNs involves challenges such as maintaining reliable communication links, ensuring data security, and addressing potential interference.

The third layer of the framework includes data collection and data processing methods that are essential for transforming raw data into actionable insights. UAVs offer significant advantages over traditional data collection methods, such as "in situ" technologies and floating car data (FCD), in terms of flexibility, cost-effectiveness, and coverage. UAVs can capture high-resolution traffic data from various perspectives, providing valuable information for analysis. Researchers have developed various data processing techniques for extracting traffic parameters from UAV-collected video data. These extracted traffic parameters can be used for vehicle detection, tracking, and traffic management.

Moreover, state-of-the-art research developments and the applications of UAVs in the transportation sectors are also explored. The key areas include the role of UAVs in enhancing road safety, highway infrastructure management, emergency response, delivery services, and cybersecurity. This discussion implies that UAV technology has the potential to significantly impact the transportation sector by improving safety, efficiency, and sustainability. However, addressing the challenges and implementing appropriate policies and regulations are essential for realizing the full benefits of UAV-enabled ITSs. The most significant challenges for implementing UAV-enabled ITSs include ensuring security from cyberattacks and ensuring other forms of data protection, improving energy efficiency, and developing AI-driven algorithms. These areas need further research and development to fully utilize the potential of UAV-enabled ITS.

Overall, this paper provides a comprehensive insight into the advancements and limitations of UAV technology in implementing ITSs, which will be beneficial in achieving robust ITSs in the future.

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References

1. Musa, A.A.; Malami, S.I.; Alanazi, F.; Ounaies, W.; Alshammari, M.; Haruna, S.I. Sustainable Traffic Management for Smart Cities Using Internet-of-Things-Oriented Intelligent Transportation Systems (ITS): Challenges and Recommendations. *Sustainability* **2023**, *15*, 9859. [CrossRef]
2. Alanazi, F. A systematic literature review of autonomous and connected vehicles in traffic management. *Appl. Sci.* **2023**, *13*, 1789. [CrossRef]
3. Curri, A. Traffic Management with Intelligent Transportation Systems. *J. Traffic Transp. Manag.* **2023**, *4*, 71–74.
4. Menouar, H.; Guvenc, I.; Akkaya, K.; Uluagac, A.S.; Kadri, A.; Tuncer, A. UAV-enabled intelligent transportation systems for the smart city: Applications and challenges. *IEEE Commun. Mag.* **2017**, *55*, 22–28. [CrossRef]
5. Saboor, A.; Coene, S.; Vinogradov, E.; Tanghe, E.; Joseph, W.; Pollin, S. Elevating the future of mobility: UAV-enabled Intelligent Transportation Systems. *arXiv* **2021**, arXiv:2110.09934.
6. Khan, N.A.; Jhanjhi, N.; Brohi, S.N.; Usmani, R.S.A.; Nayyar, A. Smart traffic monitoring system using unmanned aerial vehicles (UAVs). *Comput. Commun.* **2020**, *157*, 434–443. [CrossRef]
7. Afrin, T.; Yodo, N. A probabilistic estimation of traffic congestion using Bayesian network. *Measurement* **2021**, *174*, 109051.
8. Pishue, B. 2020 Global Traffic Scorecard. 2020. Available online: https://www.valoraanalitik.com/wp-content/uploads/2021/03/2020_INRIX_Scorecard_Report_US.pdf (accessed on 1 August 2024).
9. Pishue, B. 2022 Global Traffic Scorecard. INRIX, Inc. 2023. Available online: https://lesvoitures.fr/wp-content/uploads/2023/01/2022_INRIX_Traffic_Scorecard_Report.pdf (accessed on 1 August 2024).
10. Iyer, L.S. AI enabled applications towards intelligent transportation. *Transp. Eng.* **2021**, *5*, 100083. [CrossRef]
11. Gupta, A.; Afrin, T.; Scully, E.; Yodo, N. Advances of UAVs toward future transportation: The state-of-the-art, challenges, and opportunities. *Future Transp.* **2021**, *1*, 326–350. [CrossRef]
12. Elloumi, M.; Dhaou, R.; Escrig, B.; Idoudi, H.; Saidane, L.A. Monitoring road traffic with a UAV-based system. In Proceedings of the 2018 IEEE Wireless Communications and Networking Conference (WCNC), Barcelona, Spain, 15–18 April 2018.
13. Lucic, M.C.; Bouhamed, O.; Ghazzai, H.; Khanfor, A.; Massoud, Y. Leveraging UAVs to enable dynamic and smart aerial infrastructure for ITS and smart cities: An overview. *Drones* **2023**, *7*, 79. [CrossRef]
14. Fotouhi, A.; Qiang, H.; Ding, M.; Hassan, M.; Giordano, L.G.; Garcia-Rodriguez, A.; Yuan, J. Survey on UAV cellular communications: Practical aspects, standardization advancements, regulation, and security challenges. *IEEE Commun. Surv. Tutor.* **2019**, *21*, 3417–3442. [CrossRef]
15. Zhi, Y.; Fu, Z.; Sun, X.; Yu, J. Security and privacy issues of UAV: A survey. *Mob. Netw. Appl.* **2019**, *25*, 95–101. [CrossRef]
16. Raiyn, J. Detection of road traffic anomalies based on computational data science. *Discov. Internet Things* **2022**, *2*, 6. [CrossRef]
17. Choi, H.-W.; Kim, H.-J.; Kim, S.-K.; Na, W.S. An overview of drone applications in the construction industry. *Drones* **2023**, *7*, 515. [CrossRef]
18. Adoni, W.Y.H.; Lorenz, S.; Fareedh, J.S.; Gloaguen, R.; Bussmann, M. Investigation of autonomous multi-UAV systems for target detection in distributed environment: Current developments and open challenges. *Drones* **2023**, *7*, 263. [CrossRef]
19. Guerrero-Sánchez, M.-E.; Hernández-González, O.; Lozano, R.; García-Beltrán, C.-D.; Valencia-Palomo, G.; López-Estrada, F.-R. Energy-based control and LMI-based control for a quadrotor transporting a payload. *Mathematics* **2019**, *7*, 1090. [CrossRef]
20. Afrin, T.; Yodo, N. Reliability Analysis of Unmanned Aerial Vehicle Communication Networks (UAVCNs) for Intelligent Transportation Systems based on Energy Consumptions. In Proceedings of the Institute of Industrial and Systems Engineers (IISE) Annual Conference and Expo 2022, Seattle, WA, USA, 21–24 May 2022.
21. Nagel, L. A Guide to Lithium Polymer Batteries for Drones. 2021. Available online: https://www.tytorobotics.com/blogs/articles/a-guide-to-lithium-polymer-batteries-for-drones (accessed on 1 August 2024).
22. Boukoberine, M.N.; Zhou, Z.; Benbouzid, M. A critical review on unmanned aerial vehicles power supply and energy management: Solutions, strategies, and prospects. *Appl. Energy* **2019**, *255*, 113823. [CrossRef]
23. Xiao, C.; Wang, B.; Zhao, D.; Wang, C. Comprehensive investigation on Lithium batteries for electric and hybrid-electric unmanned aerial vehicle applications. *Therm. Sci. Eng. Prog.* **2023**, *38*, 101677. [CrossRef]
24. Alhawi, O. Finding Software Vulnerabilities in Unmanned Aerial Vehicles. Ph.D. Thesis, University of Manchester, Manchester, UK, 2021.
25. Jackson, B. Top 5 Drone Mapping Software for 2024. 2024. Available online: https://coptrz.com/top-5-drone-mapping-software/ (accessed on 1 August 2024).
26. Koyuncu, E.; Shabanighazikelayeh, M.; Seferoglu, H. Deployment and trajectory optimization of UAVs: A quantization theory approach. *IEEE Trans. Wirel. Commun.* **2018**, *17*, 8531–8546. [CrossRef]
27. Guerrero-Sanchez, M.-E.; Hernandez-Gonzalez, O.; Valencia-Palomo, G.; Lopez-Estrada, F.-R.; Rodriguez-Mata, A.-E.; Garrido, J. Filtered observer-based ida-pbc control for trajectory tracking of a quadrotor. *IEEE Access* **2021**, *9*, 114821–114835. [CrossRef]

28. Dai, H.; Zhang, H.; Wang, B.; Yang, L. The multi-objective deployment optimization of UAV-mounted cache-enabled base stations. *Phys. Commun.* **2019**, *34*, 114–120. [CrossRef]
29. Brust, M.R.; Strimbu, B.M. A networked swarm model for uav deployment in the assessment of forest environments. In Proceedings of the 2015 IEEE Tenth International Conference on Intelligent Sensors, Sensor Networks and Information Processing (ISSNIP), Singapore, 7–9 April 2015.
30. Zhang, X.; Duan, L. Optimization of emergency UAV deployment for providing wireless coverage. In Proceedings of the GLOBECOM 2017-2017 IEEE Global Communications Conference, Singapore, 4–8 December 2017.
31. Ruan, L.; Wang, J.; Chen, J.; Xu, Y.; Yang, Y.; Jiang, H.; Zhang, Y.; Xu, Y. Energy-efficient multi-UAV coverage deployment in UAV networks: A game-theoretic framework. *China Commun.* **2018**, *15*, 194–209. [CrossRef]
32. Zhao, H.; Wang, H.; Wu, W.; Wei, J. Deployment algorithms for UAV airborne networks toward on-demand coverage. *IEEE J. Sel. Areas Commun.* **2018**, *36*, 2015–2031. [CrossRef]
33. Wang, Z.; Duan, L.; Zhang, R. Adaptive deployment for UAV-aided communication networks. *IEEE Trans. Wirel. Commun.* **2019**, *18*, 4531–4543. [CrossRef]
34. Liu, X.; Liu, Y.; Chen, Y. Reinforcement learning in multiple-UAV networks: Deployment and movement design. *IEEE Trans. Veh. Technol.* **2019**, *68*, 8036–8049. [CrossRef]
35. Fu, S.; Tang, Y.; Zhang, N.; Zhao, L.; Wu, S.; Jian, X. Joint unmanned aerial vehicle (UAV) deployment and power control for Internet of Things networks. *IEEE Trans. Veh. Technol.* **2020**, *69*, 4367–4378. [CrossRef]
36. Valiulahi, I.; Masouros, C. Multi-UAV Deployment for Throughput Maximization in the Presence of Co-Channel Interference. *IEEE Internet Things J.* **2020**, *8*, 3605–3618. [CrossRef]
37. Huang, H.; Savkin, A.V.; Huang, C. Decentralised Autonomous Navigation of a UAV Network for Road Traffic Monitoring. *IEEE Trans. Aerosp. Electron. Syst.* **2021**, *57*, 2558–2564. [CrossRef]
38. Alzenad, M.; El-Keyi, A.; Lagum, F.; Yanikomeroglu, H. 3-D placement of an unmanned aerial vehicle base station (UAV-BS) for energy-efficient maximal coverage. *IEEE Wirel. Commun. Lett.* **2017**, *6*, 434–437. [CrossRef]
39. Mozaffari, M.; Saad, W.; Bennis, M.; Debbah, M. Wireless communication using unmanned aerial vehicles (UAVs): Optimal transport theory for hover time optimization. *IEEE Trans. Wirel. Commun.* **2017**, *16*, 8052–8066. [CrossRef]
40. Koyuncu, E. Power-efficient deployment of UAVs as relays. In Proceedings of the 2018 IEEE 19th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC), Kalamata, Greece, 25–28 June 2018.
41. Kyrkou, C.; Timotheou, S.; Kolios, P.; Theocharides, T.; Panayiotou, C.G. Optimized vision-directed deployment of UAVs for rapid traffic monitoring. In Proceedings of the 2018 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 12–14 January 2018.
42. Kuru, K. Planning the future of smart cities with swarms of fully autonomous unmanned aerial vehicles using a novel framework. *IEEE Access* **2021**, *9*, 6571–6595. [CrossRef]
43. Reina, D.; Tawfik, H.; Toral, S. Multi-subpopulation evolutionary algorithms for coverage deployment of UAV-networks. *Ad Hoc Netw.* **2018**, *68*, 16–32. [CrossRef]
44. Fabra, F.; Vegni, A.M.; Loscrì, V.; Calafate, C.T.; Cano, J.; Manzoni, P. Collision-free Cooperative UAV Protocols for Sustainable Aerial Services. *IET Smart Cities* **2022**. Available online: https://hal.science/hal-03616495/ (accessed on 1 August 2024). [CrossRef]
45. Garg, P.K. Potentials of Network-Based Unmanned Aerial Vehicles. In *Cloud and IoT-Based Vehicular Ad Hoc Networks*; Scrivener Publishing: Beverly, MA, USA, 2021; pp. 369–397.
46. Hentati, A.I.; Fourati, L.C. Comprehensive survey of UAVs communication networks. *Comput. Stand. Interfaces* **2020**, *72*, 103451. [CrossRef]
47. Gupta, L.; Jain, R.; Vaszkun, G. Survey of important issues in UAV communication networks. *IEEE Commun. Surv. Tutor.* **2016**, *18*, 1123–1152. [CrossRef]
48. Khan, M.A.; Ectors, W.; Bellemans, T.; Janssens, D.; Wets, G. UAV-based traffic analysis: A universal guiding framework based on literature survey. *Transp. Res. Procedia* **2017**, *22*, 541–550. [CrossRef]
49. Barmpounakis, E.N.; Vlahogianni, E.I.; Golias, J.C. Unmanned Aerial Aircraft Systems for transportation engineering: Current practice and future challenges. *Int. J. Transp. Sci. Technol.* **2016**, *5*, 111–122. [CrossRef]
50. Barmpounakis, E.; Geroliminis, N. On the new era of urban traffic monitoring with massive drone data: The pNEUMA large-scale field experiment. *Transp. Res. Part C Emerg. Technol.* **2019**, *111*, 50–71. [CrossRef]
51. Niu, H.; Gonzalez-Prelcic, N.; Heath, R.W. A UAV-based traffic monitoring system-invited paper. In Proceedings of the 2018 IEEE 87th Vehicular Technology Conference (VTC Spring), Porto, Portugal, 3–6 June 2018.
52. Li, M.; Zhen, L.; Wang, S.; Lv, W.; Qu, X. Unmanned aerial vehicle scheduling problem for traffic monitoring. *Comput. Ind. Eng.* **2018**, *122*, 15–23. [CrossRef]
53. de Frías, C.J.; Al-Kaff, A.; Moreno, F.M.; Madridano, Á.; Armingol, J.M. Intelligent Cooperative System for Traffic Monitoring in Smart Cities. In Proceedings of the 2020 IEEE Intelligent Vehicles Symposium (IV), Las Vegas, NV, USA, 19 October–13 November 2020.
54. Choi, K.; Lee, I. A UAV-based close-range rapid aerial monitoring system for emergency responses. In Proceedings of the ISPRS Zurich 2011 Workshop, Zurich, Switzerland, 14–16 September 2011; International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences. Volume XXXVIII-1/C22, pp. 247–252.

55. Doherty, P.; Granlund, G.; Kuchcinski, K.; Sandewall, E.; Nordberg, K.; Skarman, E.; Wiklund, J. The WITAS unmanned aerial vehicle project. In Proceedings of the 14th European Conference on Artificial Intelligence (ECAI), Berlin, Germany, 20–25 August 2000; pp. 747–755.
56. Nordberg, K.; Doherty, P.; Farnebäck, G.; Forssén, P.E.; Granlund, G.; Moe, A.; Wiklund, J. Vision for a UAV helicopter. In Proceedings of the International Conference on Intelligent Robots and Systems (IROS), Lausanne, Switzerland, 30 September–4 October 2002.
57. Rasmussen, S.; Kalyanam, K.; Kingston, D. Field experiment of a fully autonomous multiple UAV/UGS intruder detection and monitoring system. In Proceedings of the 2016 International Conference on Unmanned Aircraft Systems (ICUAS), Arlington, VA, USA, 7–10 June 2016.
58. Rasmussen, S.; Kingston, D.; Humphrey, L. A brief introduction to unmanned systems autonomy services (UxAS). In Proceedings of the 2018 International Conference on Unmanned Aircraft Systems (ICUAS), Dallas, TX, USA, 2–15 June 2018.
59. Du, Y.; Zhao, C.; Li, F.; Yang, X. An open data platform for traffic parameters measurement via multirotor unmanned aerial vehicles video. *J. Adv. Transp.* **2017**, *2017*, 8324301. [CrossRef]
60. Ke, R.; Feng, S.; Cui, Z.; Wang, Y. Advanced framework for microscopic and lane-level macroscopic traffic parameters estimation from UAV video. *IET Intell. Transp. Syst.* **2020**, *14*, 724–734. [CrossRef]
61. Ke, R.; Li, Z.; Tang, J.; Pan, Z.; Wang, Y. Real-time traffic flow parameter estimation from UAV video based on ensemble classifier and optical flow. *IEEE Trans. Intell. Transp. Syst.* **2019**, *20*, 54–64. [CrossRef]
62. Ke, R.; Li, Z.; Kim, S.; Ash, J.; Cui, Z.; Wang, Y. Real-time bidirectional traffic flow parameter estimation from aerial videos. *IEEE Trans. Intell. Transp. Syst.* **2016**, *18*, 890–901. [CrossRef]
63. Brkić, I.; Miler, M.; Ševrović, M.; Medak, D. An Analytical Framework for Accurate Traffic Flow Parameter Calculation from UAV Aerial Videos. *Remote Sens.* **2020**, *12*, 3844. [CrossRef]
64. Leduc, G. Road traffic data: Collection methods and applications. *Work. Pap. Energy Transp. Clim. Change* **2008**, *1*, 1–55.
65. Zhu, J.; Sun, K.; Jia, S.; Li, Q.; Hou, X.; Lin, W.; Liu, B.; Qiu, G. Urban traffic density estimation based on ultrahigh-resolution UAV video and deep neural network. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2018**, *11*, 4968–4981. [CrossRef]
66. Khan, M.A.; Ectors, W.; Bellemans, T.; Janssens, D.; Wets, G. Unmanned aerial vehicle–based traffic analysis: Methodological framework for automated multivehicle trajectory extraction. *Transp. Res. Rec. J. Transp. Res. Board* **2017**, *2626*, 25–33. [CrossRef]
67. Khan, M.A.; Ectors, W.; Bellemans, T.; Janssens, D.; Wets, G. Unmanned aerial vehicle-based traffic analysis: A case study for shockwave identification and flow parameters estimation at signalized intersections. *Remote Sens.* **2018**, *10*, 458. [CrossRef]
68. Van Lint, J.W.C.; Van Hinsbergen, C.P.I.J. Short-term traffic and travel time prediction models. *Artif. Intell. Appl. Crit. Transp. Issues* **2012**, *22*, 22–41.
69. Wu, T.; Xie, K.; Xinpin, D.; Song, G. A online boosting approach for traffic flow forecasting under abnormal conditions. In Proceedings of the 2012 9th International Conference on Fuzzy Systems and Knowledge Discovery (FSKD), Chongqing, China, 29-31 May 2012.
70. Shepelev, V.; Aliukov, S.; Nikolskaya, K.; Shabiev, S. The capacity of the road network: Data collection and statistical analysis of traffic characteristics. *Energies* **2020**, *13*, 1765. [CrossRef]
71. Zhao, Z.; Chen, W.; Wu, X.; Chen, P.C.Y.; Liu, J. LSTM network: A deep learning approach for short-term traffic forecast. *IET Intell. Transp. Syst.* **2017**, *11*, 68–75. [CrossRef]
72. Afrin, T.; Yodo, N. Correlated Traffic Data Prediction Towards Robust Traffic Management System. In Proceedings of the Institute of Industrial and Systems Engineers (IISE) Annual Conference, Online, 22–25 May 2021.
73. Koesdwiady, A.; Soua, R.; Karray, F. Improving traffic flow prediction with weather information in connected cars: A deep learning approach. *IEEE Trans. Veh. Technol.* **2016**, *65*, 9508–9517. [CrossRef]
74. Guo, J.; Huang, W.; Williams, B.M. Adaptive Kalman filter approach for stochastic short-term traffic flow rate prediction and uncertainty quantification. *Transp. Res. Part C Emerg. Technol.* **2014**, *43*, 50–64. [CrossRef]
75. Eckstein, W.; Steger, C. The Halcon vision system: An example for flexible software architecture. In Proceedings of the 3rd Japanese Conference on Practical Applications of Real-Time Image Processing, Kokura, Japan, 31 August–4 September 2014.
76. Lenz, R.; Tsai, R. Techniques for calibration of the scale factor and image center for high accuracy 3D machine vision metrology. In Proceedings of the IEEE International Conference on Robotics and Automation, Raleigh, NC, USA, 31 March–3April 1987; Volume 4, pp. 68–75.
77. Alqahtani, H.; Kumar, G. Cybersecurity in Electric and Flying Vehicles: Threats, Challenges, AI Solutions & Future Directions. *ACM Comput. Surv.* **2024**. [CrossRef]
78. Lamssaggad, A.; Benamar, N.; Hafid, A.S.; Msahli, M. A survey on the current security landscape of intelligent transportation systems. *IEEE Access* **2021**, *9*, 9180–9208. [CrossRef]
79. Yodo, N.; Goethals, P.L. Cybersecurity Management via Control Strategies for Resilient Cyber-Physical Systems. In Proceedings of the Institute of Industrial and Systems Engineers (IISE) Annual Conference, Orlando, FL, USA, 18–21 May 2019.
80. Ly, B.; Ly, R. Cybersecurity in unmanned aerial vehicles (UAVs). *J. Cyber Secur. Technol.* **2021**, *5*, 120–137. [CrossRef]
81. Shafik, W.; Matinkhah, S.M.; Shokoor, F. Cybersecurity in unmanned aerial vehicles: A review. *Int. J. Smart Sens. Intell. Syst.* **2023**, *16*, 12. [CrossRef]
82. Adil, M.; Song, H.; Mastorakis, S.; Abulkasim, H.; Farouk, A.; Jin, Z. UAV-assisted IoT applications, cybersecurity threats, AI-enabled solutions, open challenges with future research directions. *IEEE Trans. Intell. Veh.* **2023**, *9*, 4583–4605. [CrossRef]

83. United States Department of Transportation (US—DOT). The ITS JPO's New Strategic Plan for the First Half of the Decade. 2020. Available online: https://www.its.dot.gov/stratplan2020/ (accessed on 1 August 2024).
84. Liu, X.; Zou, H.; Niu, W.; Song, Y.; He, W. An approach of traffic accident scene reconstruction using unmanned aerial vehicle photogrammetry. In Proceedings of the 2019 2nd International Conference on Sensors, Signal and Image Processing, Prague, Czech, 8–10 October 2019.
85. Škorput, P.; Mandžuka, S.; Gregurić, M.; Vrančić, M.T. Applying Unmanned Aerial Vehicles (UAV) in traffic investigation process. In Proceedings of the International Conference "New Technologies, Development and Applications", Sarajevo, Bosnia and Herzegovina, 27–29 June 2019.
86. Liu, S.; Wang, S.; Shi, W.; Liu, H.; Li, Z.; Mao, T. Vehicle tracking by detection in UAV aerial video. *Sci. China Inf. Sci.* **2019**, *62*, 24101. [CrossRef]
87. Pérez, J.A.; Gonçalves, G.R.; Rangel, J.M.G.; Ortega, P.F. Accuracy and effectiveness of orthophotos obtained from low cost UASs video imagery for traffic accident scenes documentation. *Adv. Eng. Softw.* **2019**, *132*, 47–54. [CrossRef]
88. Seo, J.; Duque, L.; Wacker, J.P. Field application of UAS-based bridge inspection. *Transp. Res. Rec. J. Transp. Res. Board* **2018**, *2672*, 72–81. [CrossRef]
89. Lei, B.; Wang, N.; Xu, P.; Song, G. New crack detection method for bridge inspection using UAV incorporating image processing. *J. Aerosp. Eng.* **2018**, *31*, 04018058. [CrossRef]
90. Wu, W.; Qurishee, M.A.; Owino, J.; Fomunung, I.; Onyango, M.; Atolagbe, B. Coupling deep learning and UAV for infrastructure condition assessment automation. In Proceedings of the 2018 IEEE International Smart Cities Conference (ISC2), Kansas City, MO, USA, 16–19 September 2018.
91. Leonardi, G.; Barrile, V.; Palamara, R.; Suraci, F.; Candela, G. 3D mapping of pavement distresses using an Unmanned Aerial Vehicle (UAV) system. In Proceedings of the International Symposium on New Metropolitan Perspectives, Reggio Calabria, Italy, 22–25 May 2018.
92. Lee, J.K. UAV-Based Pothole Identification: A Photogrammetric Approach. Master's Thesis, George Mason University, Fairfax, VA, USA, 2019.
93. Zhao, N.; Lu, W.; Sheng, M.; Chen, Y.; Tang, J.; Yu, F.R.; Wong, K.-K. UAV-assisted emergency networks in disasters. *IEEE Wirel. Commun.* **2019**, *26*, 45–51. [CrossRef]
94. Beg, A.; Qureshi, A.R.; Sheltami, T.; Yasar, A. UAV-enabled intelligent traffic policing and emergency response handling system for the smart city. *Pers. Ubiquitous Comput.* **2020**, *25*, 33–50. [CrossRef]
95. Hildmann, H.; Kovacs, E. Using unmanned aerial vehicles (UAVs) as mobile sensing platforms (msps) for disaster response, civil security and public safety. *Drones* **2019**, *3*, 59. [CrossRef]
96. American Red Cross; MEASURE. *Drones for Disaster Response and Relief Operations*; American Red Cross: Washington, DC, USA, 2015.
97. Deruyck, M.; Wyckmans, J.; Joseph, W.; Martens, L. Designing UAV-aided emergency networks for large-scale disaster scenarios. *EURASIP J. Wirel. Commun. Netw.* **2018**, *2018*, 79. [CrossRef]
98. Khan, S.I.; Qadir, Z.; Munawar, H.S.; Nayak, S.R.; Budati, A.K.; Verma, K.; Prakash, D. UAVs path planning architecture for effective medical emergency response in future networks. *Phys. Commun.* **2021**, *47*, 101337. [CrossRef]
99. Marsh, I. *Drones—A View into the Future for the Logistics Sector*; Marsh: London, UK, 2015.
100. Vesel, A. Autonomous Decision Making for Air Taxi Networks. *arXiv* **2024**, arXiv:2406.14832.
101. Hegde, A.; Ghose, D.; Ajorlou, A.; Homaifar, A.; Esterline, A.; Moore, J.; Bamberger, R.; Chandler, P.; Pachter, M.; Rasmussen, S.; et al. Multi-UAV Distributed Control for Load Transportation in Precision Agriculture. In Proceedings of the AIAA Scitech 2020 Forum, Orlando, FL, USA, 6–10 January 2020.
102. Aposporis, P. A review of global and regional frameworks for the integration of an unmanned aircraft system in air traffic management. *Transp. Res. Interdiscip. Perspect.* **2024**, *24*, 101064. [CrossRef]
103. Wanyonyi Rodgers, M. Legal, Policy and Institutional Framework for Regulation of UAS in the United States of America (USA). In *The Law Regulating Unmanned Aircraft Systems in the United States, South Africa and Kenya: A Civil Aviation Perspective*; Springer: Berlin/Heidelberg, Germany, 2024; pp. 81–120.
104. Bhat, G.; Dudhedia, M.; Panchal, R.; Shirke, Y.; Angane, N.; Khonde, S.; Khedkar, S.; Pansare, J.; Bere, S.; Wahul, R.; et al. Autonomous Drones and their Influence on Standardization of Rules and Regulations for Operating—A Brief Overview. *Results Control Optim.* **2024**, *14*, 100401. [CrossRef]
105. Sharma, R.; Arya, R. UAV based long range environment monitoring system with Industry 5.0 perspectives for smart city infrastructure. *Comput. Ind. Eng.* **2022**, *168*, 108066. [CrossRef]
106. Afrin, T.; Yodo, N. A Long Short-Term Memory-based correlated traffic data prediction framework. *Knowl.-Based Syst.* **2021**, *237*, 107755. [CrossRef]
107. Singh, C.; Bekal, A.; Ashoka, A.; Shreesha, P.U.; Dsouza, L.S. Human-Drone Interaction Augmenting User Experience in Industry 5.0. In *Drone Applications for Industry 5.0*; IGI Global: Hershey, PA, USA, 2024; pp. 46–58.